



Biopotential-Based Wearable System for Real-Time Anger Detection

A Wearable EMG-ECG Device with IoT-Based Monitoring

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Problem Statement

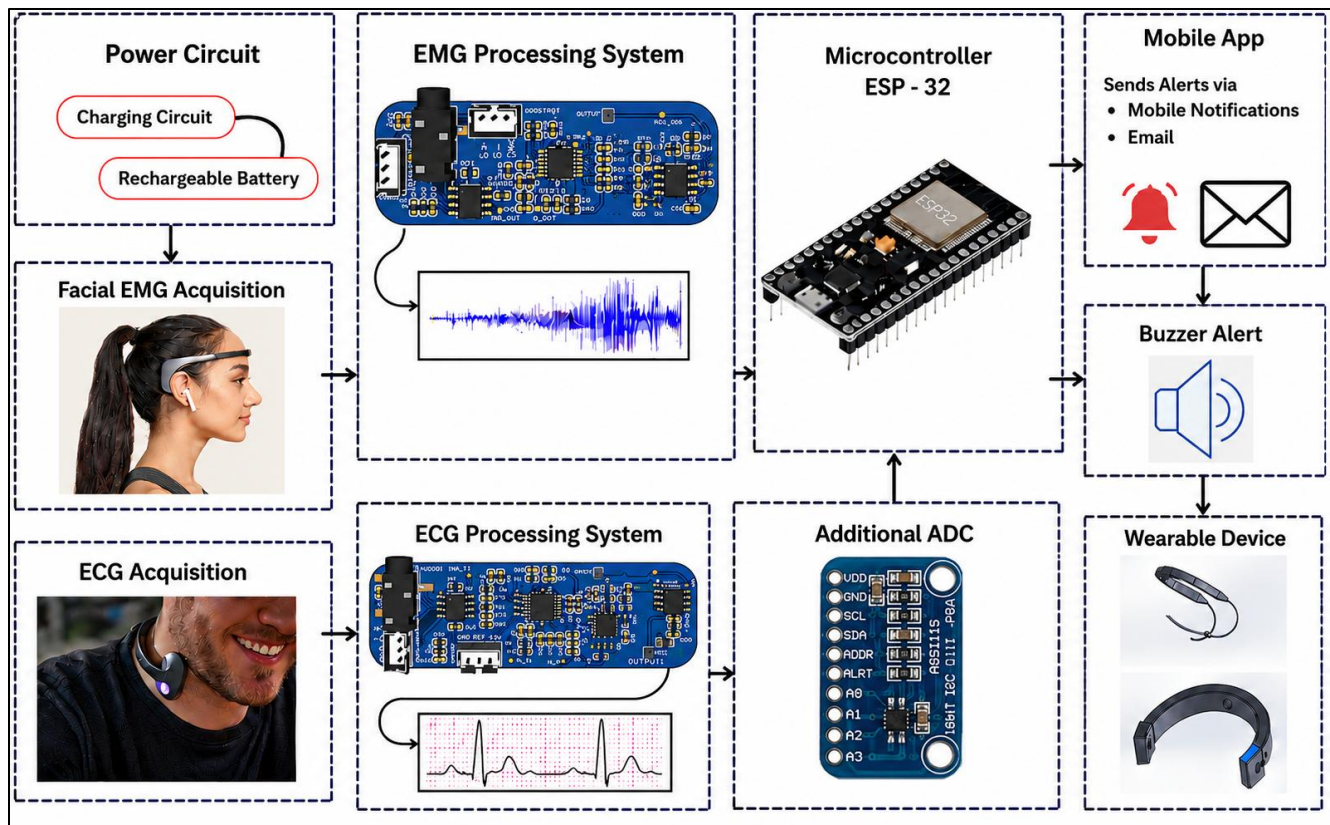
Uncontrolled anger is a major contributor to road rage, domestic violence, and workplace conflict. Most people fail to recognize the physical warning signs of rising anger, such as an increased heart rate or tensed facial muscles, until it is too late to prevent an outburst. Existing anger management methods, such as therapy and counseling, are reactive because they provide support after an episode rather than during it. There is currently no affordable wearable device that can detect anger in real time using the body's physiological signals and alert the user before the situation escalates.

Aim

To design a wearable, IoT-enabled device that detects rising anger in real time by acquiring and analyzing EMG (muscle) and ECG (heart) signals, and lays the foundation for automatic alerts before anger escalates into harmful behavior.

System Overview

The device combines two wearable bands — a forehead band for facial EMG and a neckband for ECG — connected to a microcontroller that processes and streams data to the cloud for real-time monitoring.

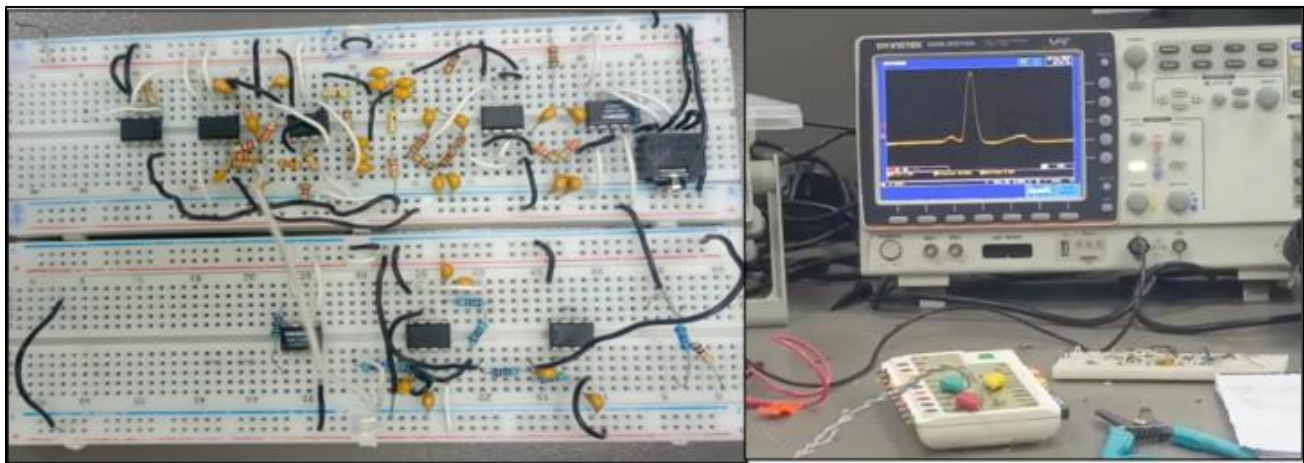


System block diagram: signal acquisition, processing, and alerting pipeline.

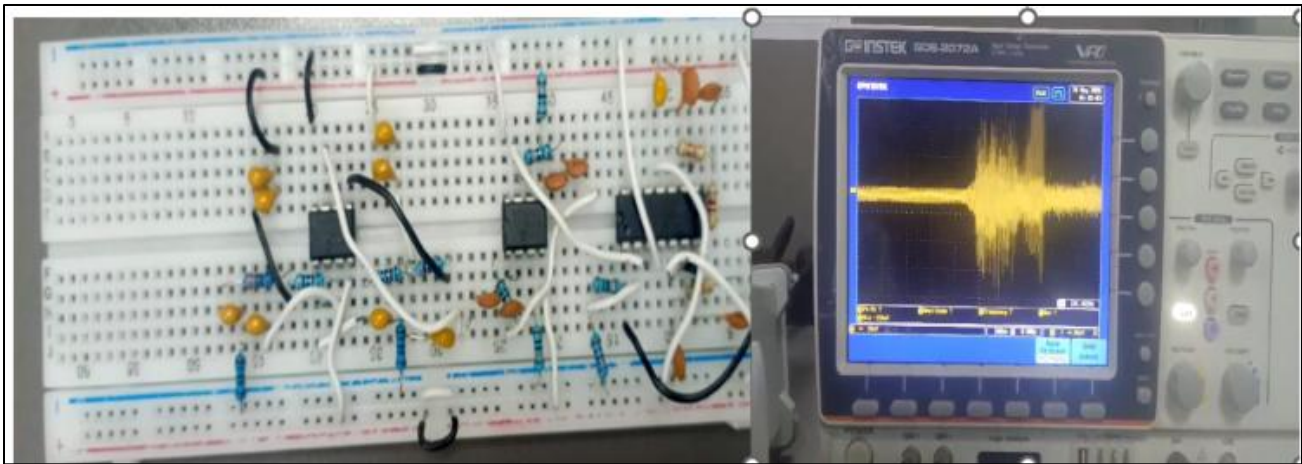
Main Development Steps

1. Signal Acquisition

Electrodes were positioned on the forehead (EMG, facial muscles) and neck (ECG). The neck was chosen for ECG so both sensors and electronics fit into one compact wearable.



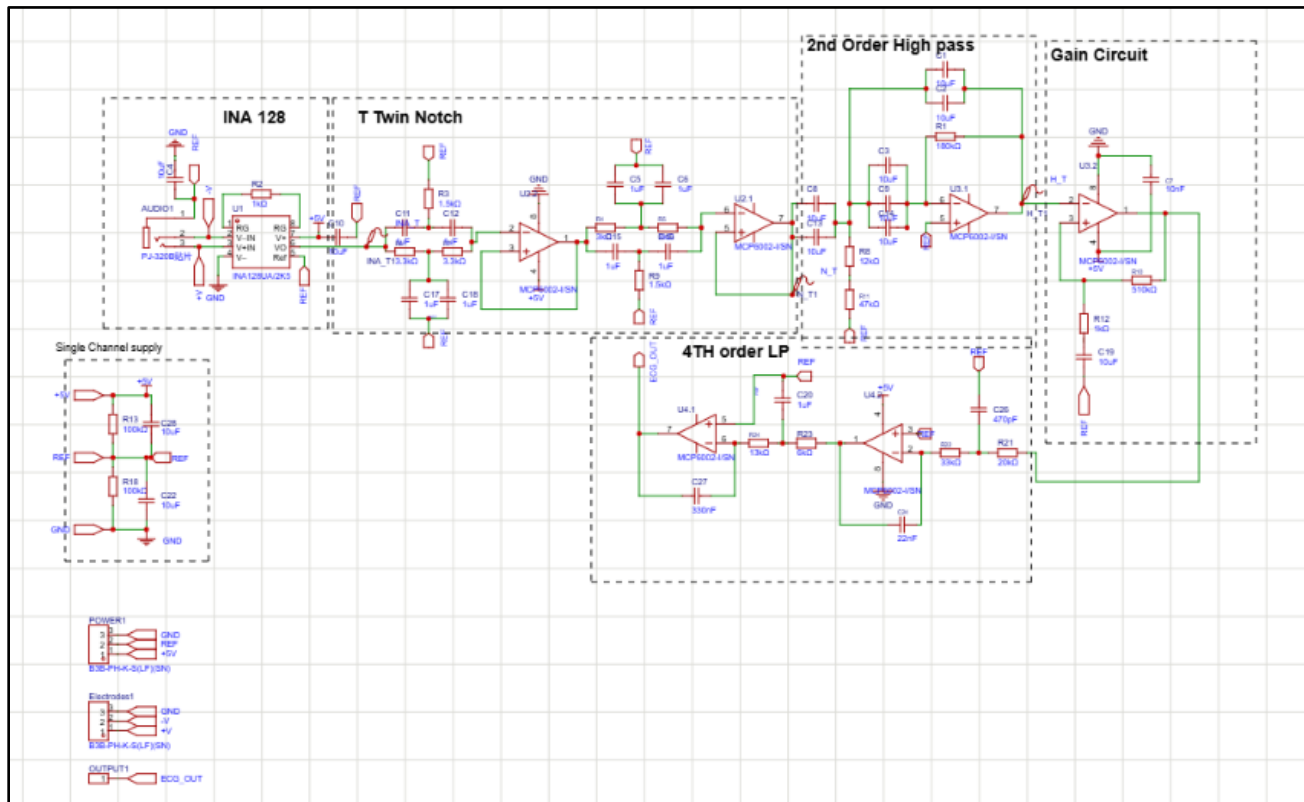
ECG acquisition testing at the neck electrode site.



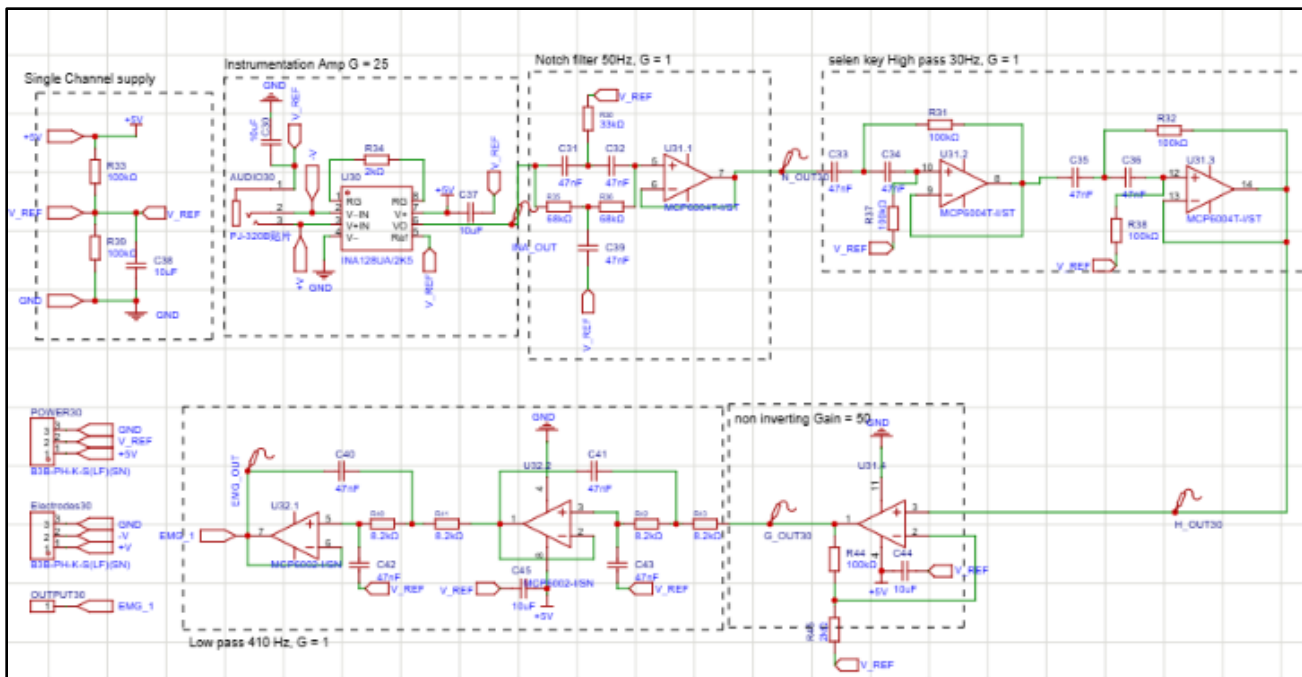
EMG acquisition testing on the forehead site, viewed on an oscilloscope.

2. Analog Front-End Circuit Design

Each channel uses an instrumentation amplifier, filters, and a notch filter to clean the raw signal and remove electrical noise before digitizing it.



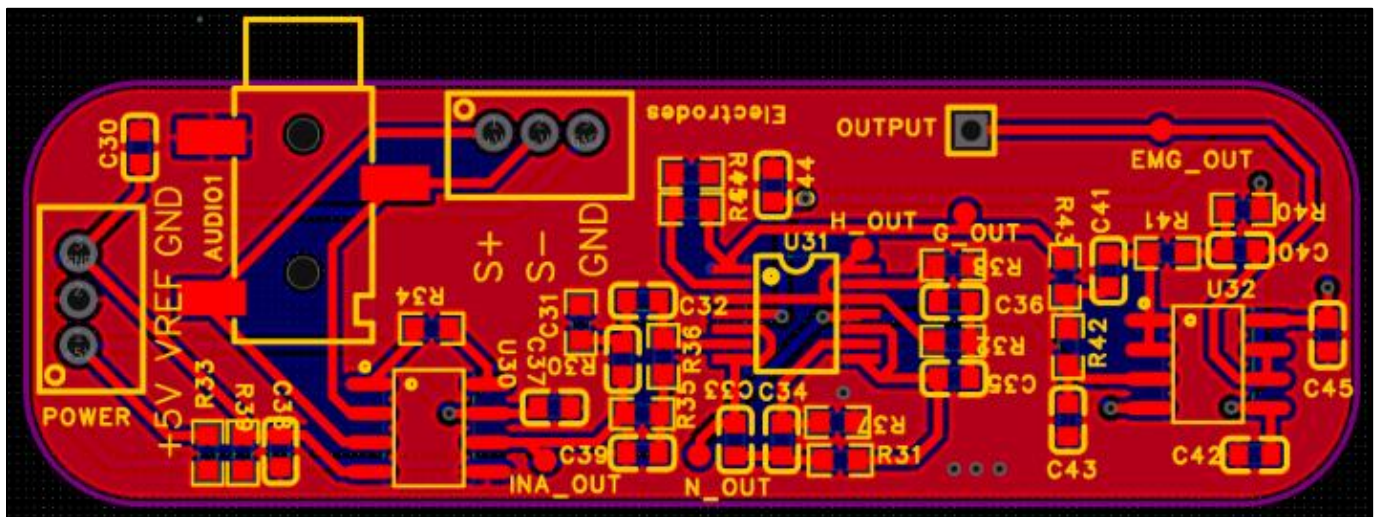
ECG signal-conditioning circuit schematic

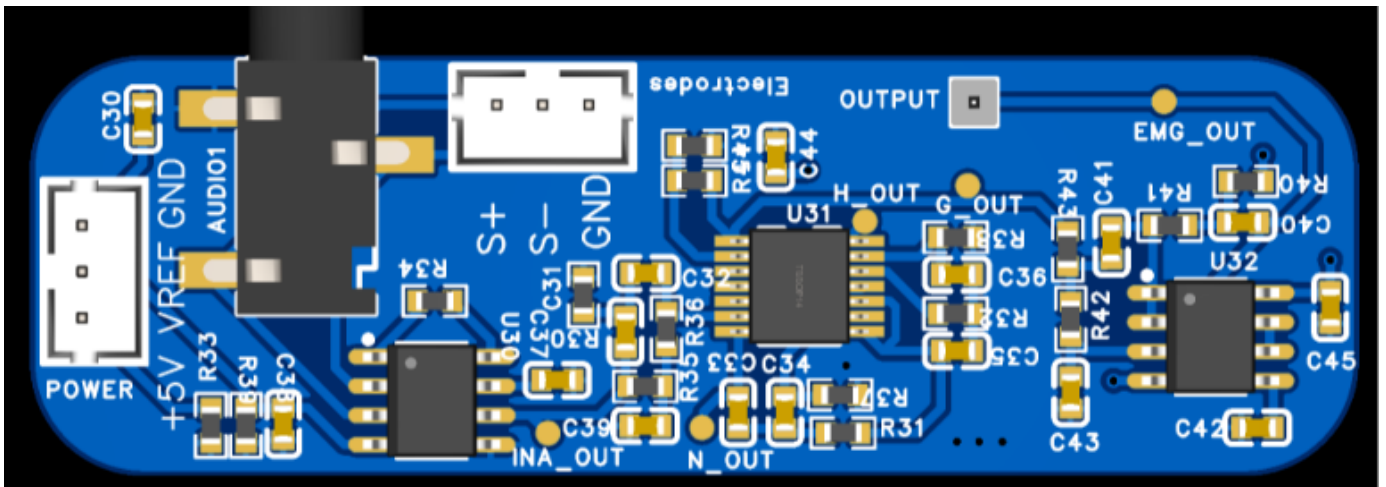


EMG signal-conditioning circuit schematic

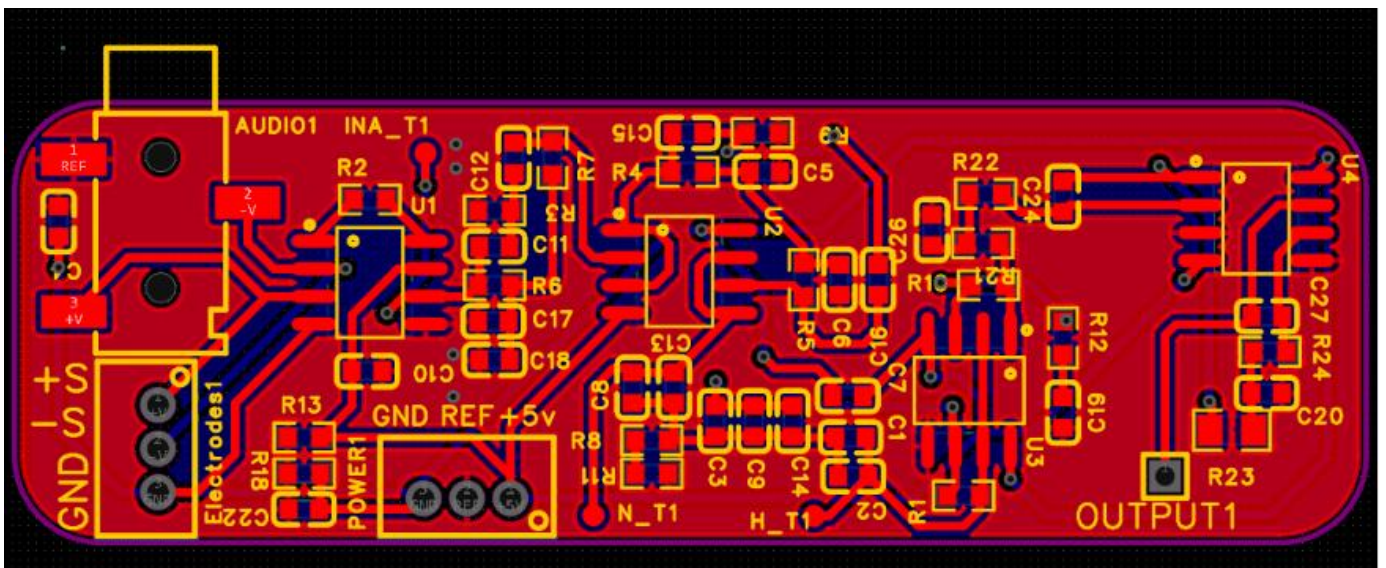
3. PCB Design & Fabrication

After breadboard testing, dedicated PCBs were designed for the EMG and ECG channels and fabricated through JLCPCB, keeping the electronics compact enough for the wearable bands.

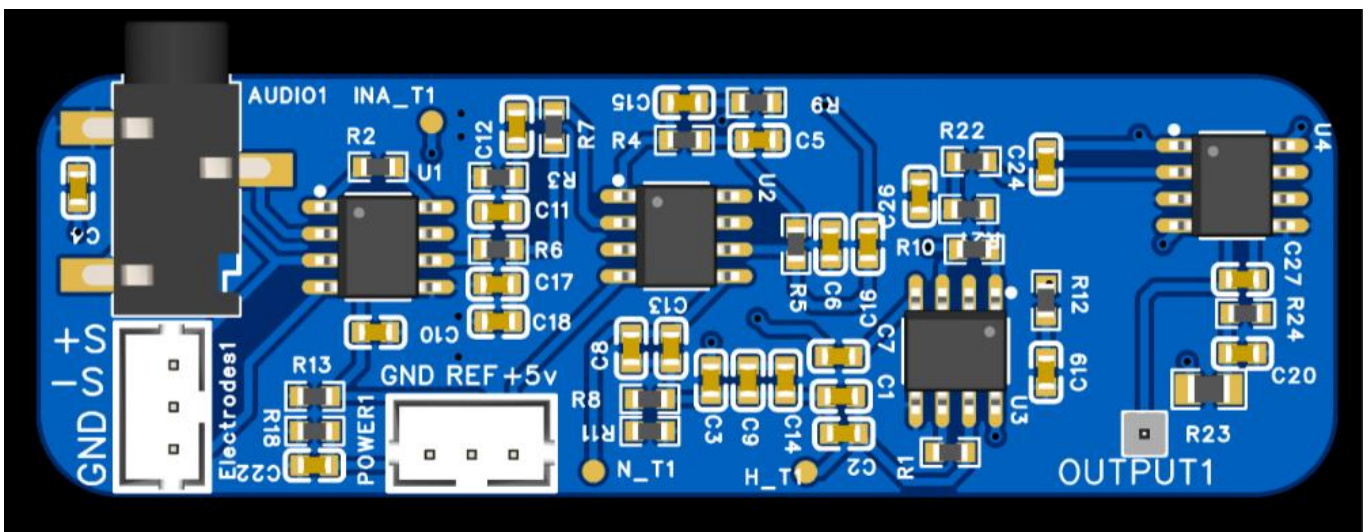




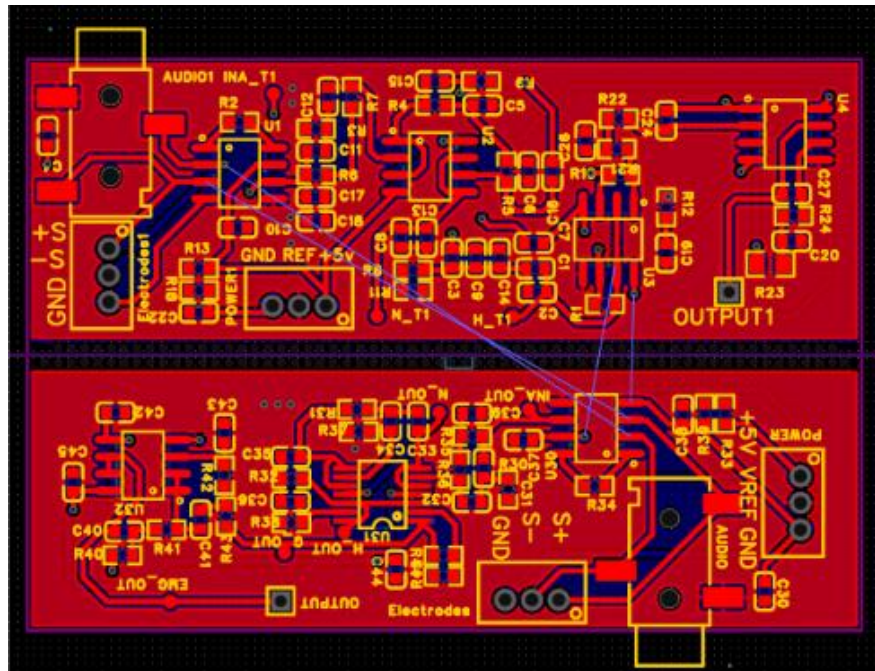
3D View EMG PCB



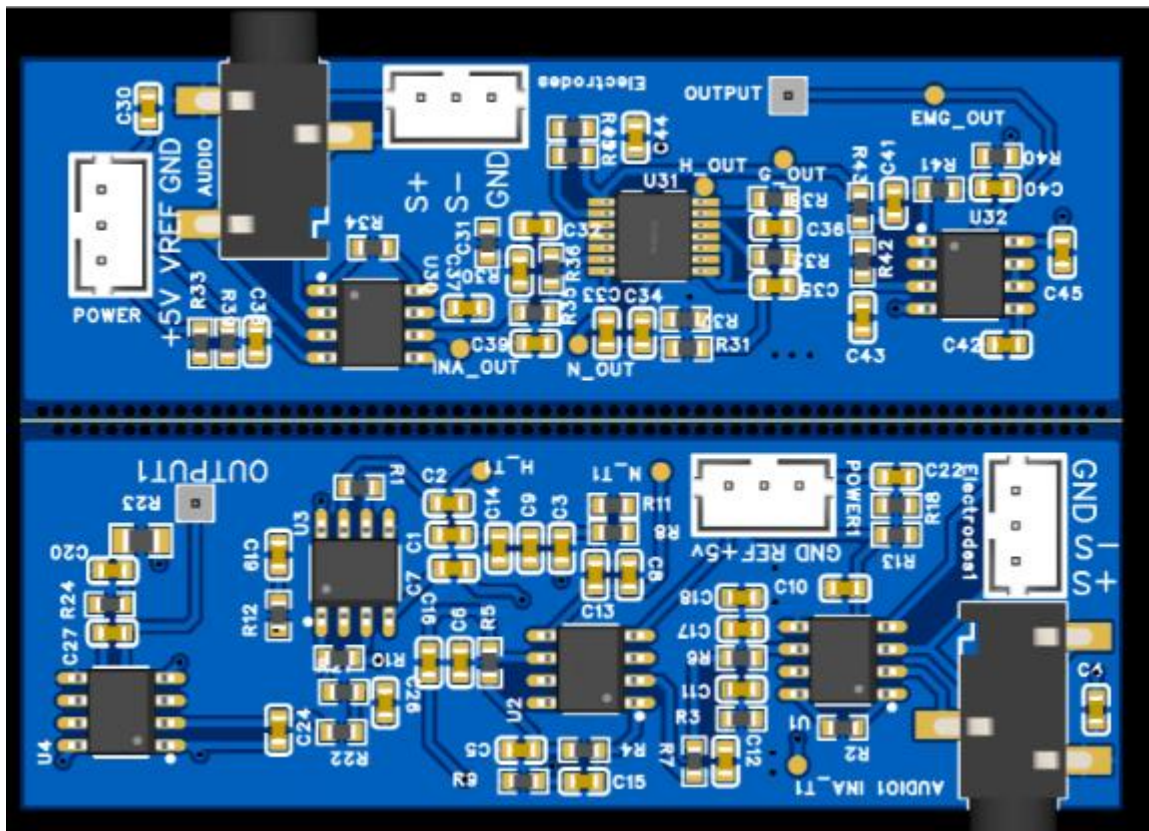
fabricated ECG PCB



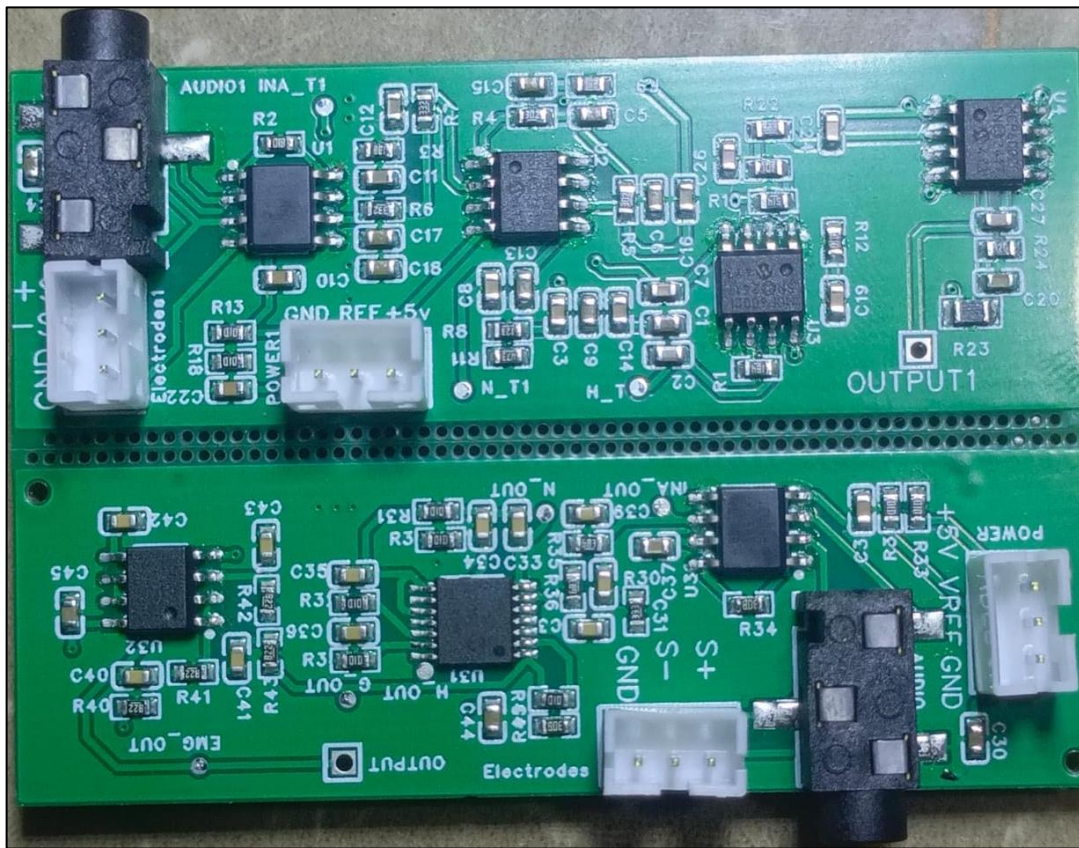
3D View ECG PCB



Combined fabricated EMG and ECG PCBs on single Board



Combined 3D View of EMG and ECG PCBs on single Board



Assembled PCB of EMG and ECG on single Board

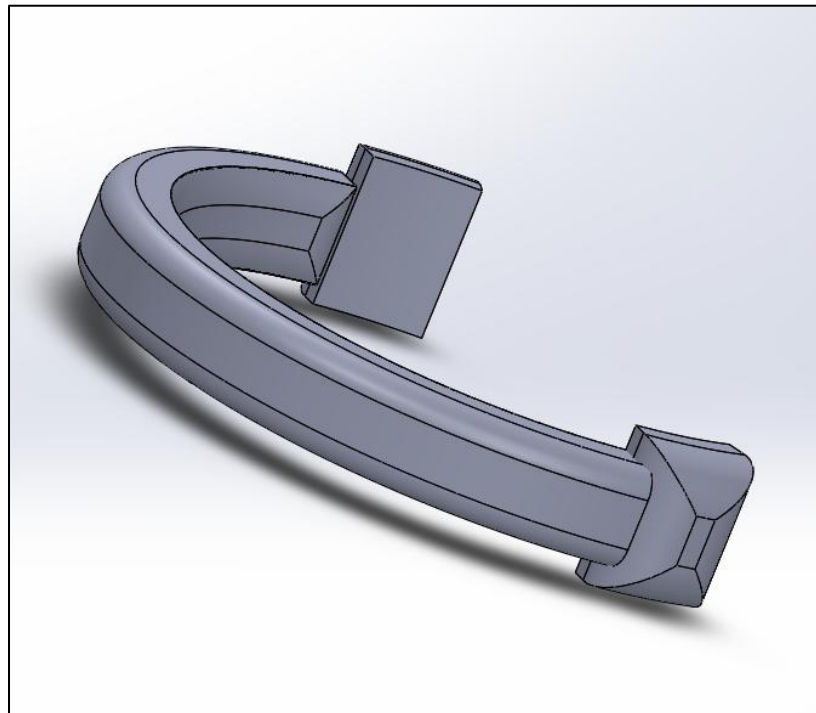




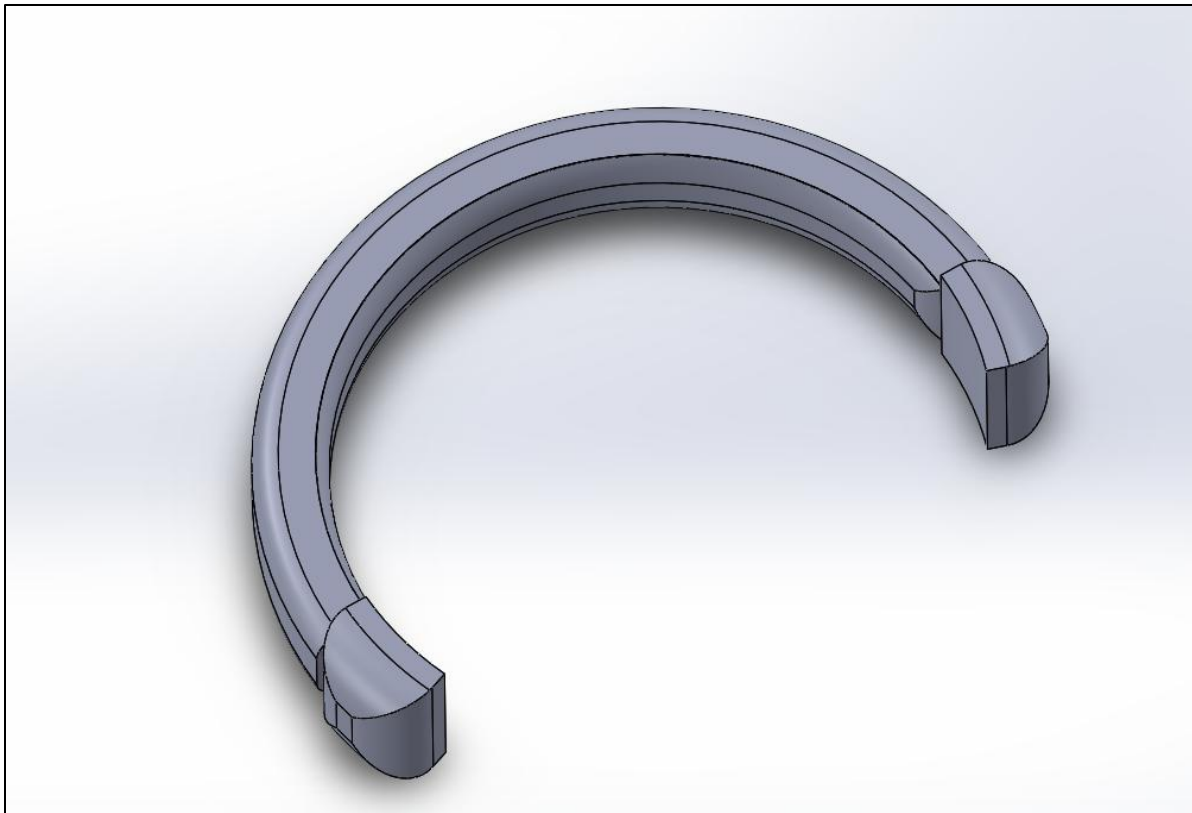
Assembled PCB of EMG and ECG in Compact Size

4. Wearable Enclosure Design

A neckband and forehead band were modeled in SolidWorks and 3D-printed to hold the electrodes and electronics comfortably in place during wear.



3D CAD model of the wearable band design.



CAD rendering of the neckband.

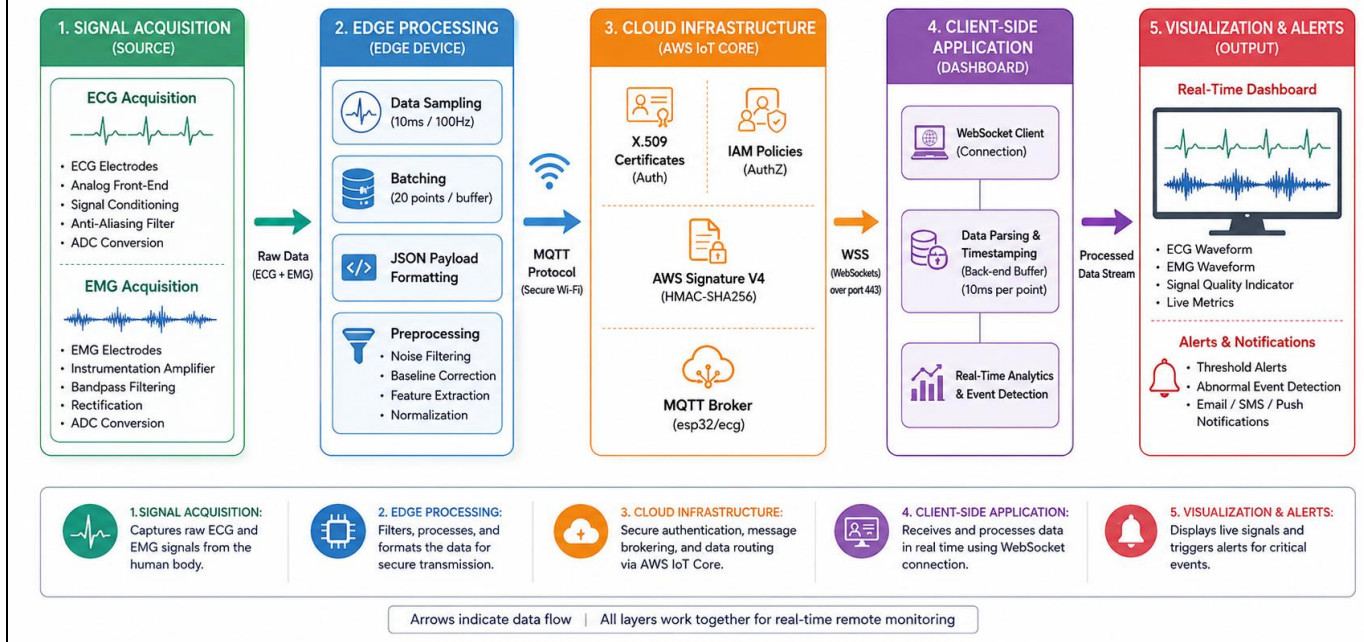


CAD rendering of the headband

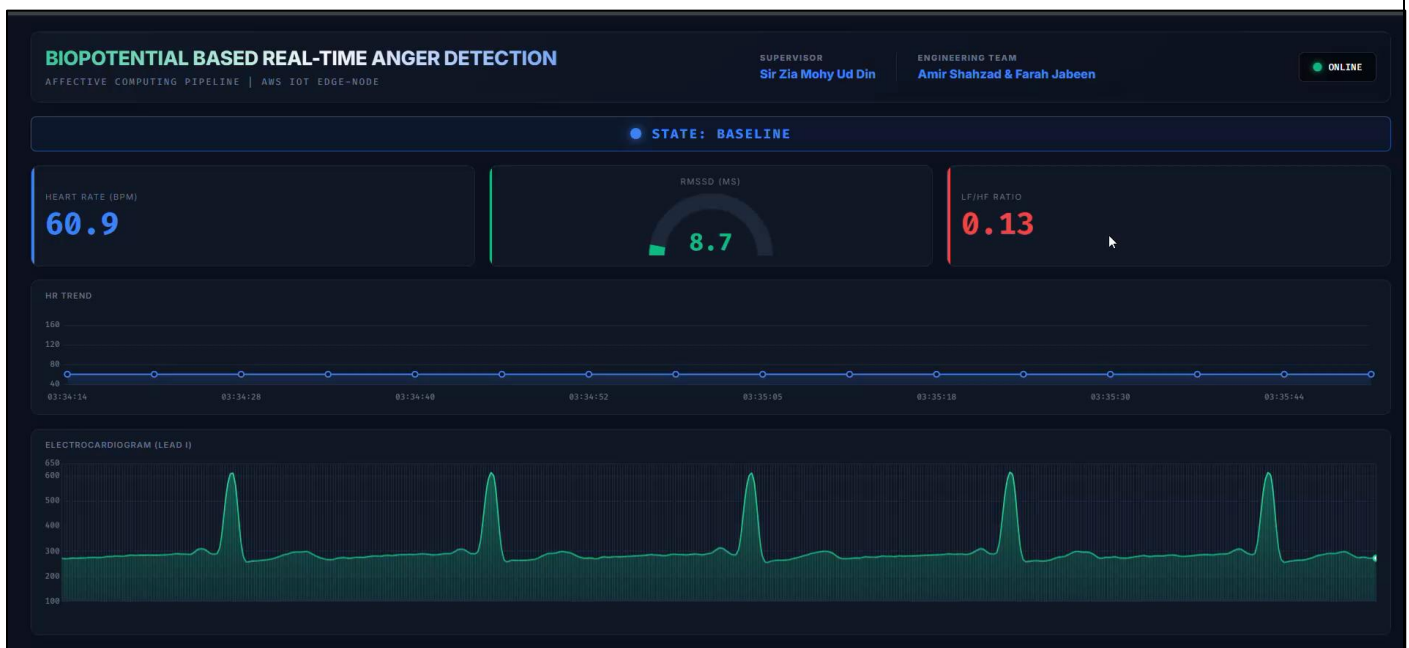
5. IoT Integration & Real-Time Dashboard

The microcontroller streams signal data over Wi-Fi to an AWS-based cloud dashboard, enabling real-time waveform visualization as the foundation for future automatic alerts.

IoT Biomedical Monitoring System - High-Level Block Diagram



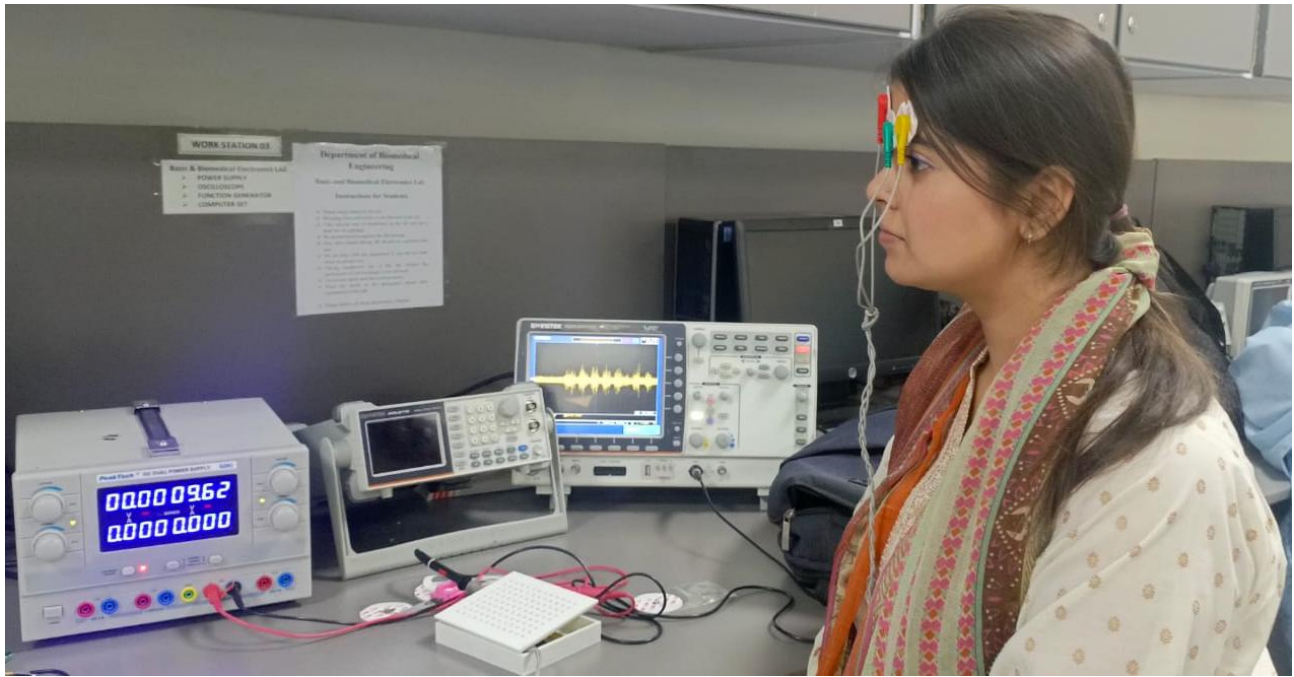
IoT architecture: from signal acquisition to cloud dashboard.



Live ECG waveform streamed to the real-time cloud dashboard.

6. Testing & Validation

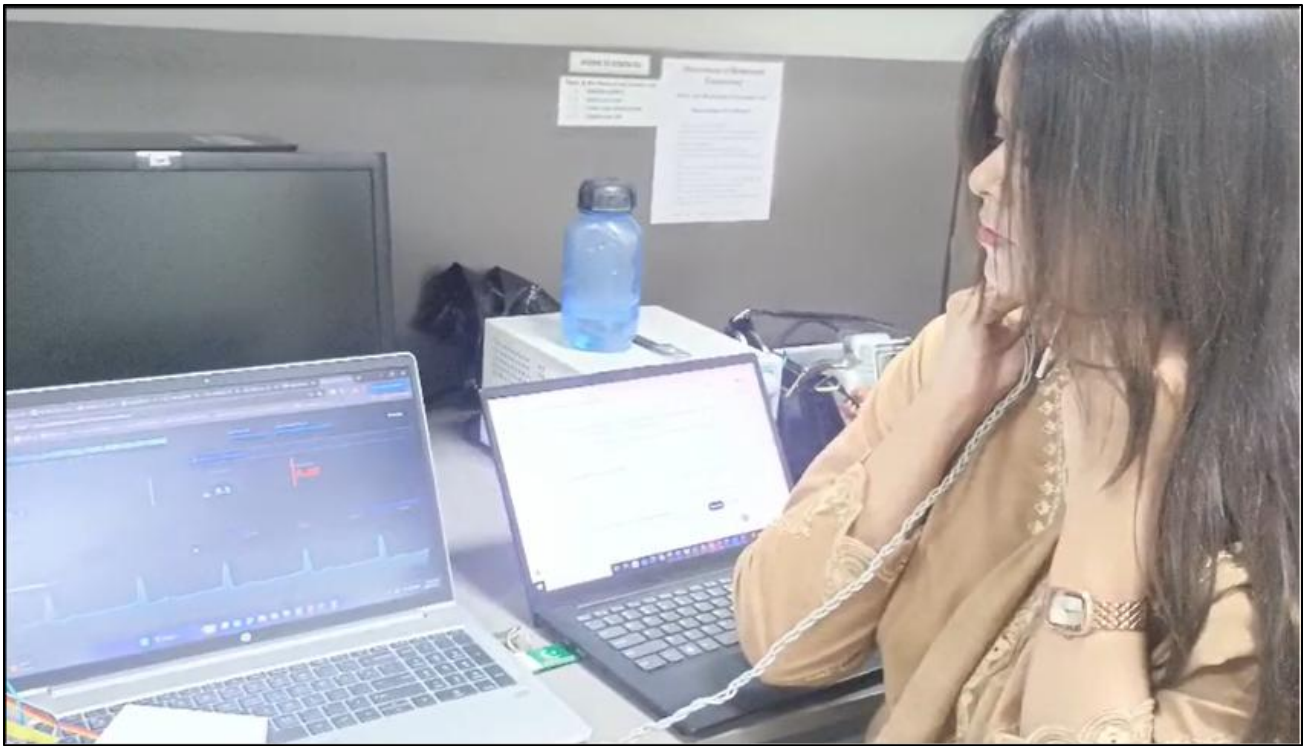
The acquired EMG and ECG signals were compared against a BIOPAC reference system to confirm accuracy and signal quality.



Facial EMG comparison test using our Printed PCB.

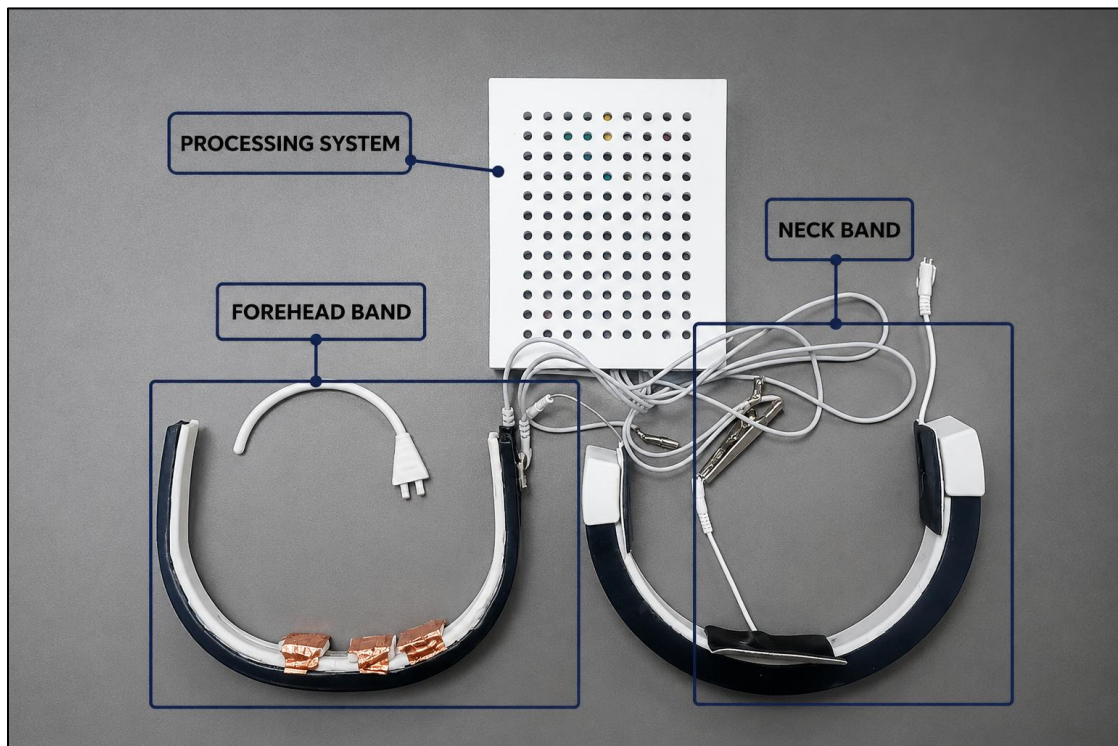


Facial EMG comparison test using the BIOPAC reference system.



Final Device

The completed prototype integrates the EMG forehead band, ECG neckband, and processing electronics into a single wearable system.



Final assembled wearable device: forehead band, neckband, and processing unit.

Applications

- Personal anger and stress self-awareness
- Driver monitoring to reduce road-rage-related incidents
- Workplace and occupational stress monitoring
- Support tool for anger-management therapy

Conclusion

This project delivers a working wearable EMG-ECG acquisition system with real-time cloud connectivity, validated against a clinical reference system. It lays the groundwork for a complete closed-loop anger-detection and alerting device through future machine learning integration.